Error in regression models

Frank Edwards

Refresher on regression

The linear regression model

Let's take a look at the regression equation

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

The linear regression model

Let's take a look at the regression equation

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

β_0: The value of y when x is equal to zero

The linear regression model

Let's take a look at the regression equation

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

β_0: The value of y when x is equal to zero

β_1: The average increase in y when x increases by one unit

The linear regression model

Let's take a look at the regression equation

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

β_0: The value of y when x is equal to zero

β_1: The average increase in y when x increases by one unit

ε: The distance between the line $y = \beta_0 + \beta_1 X$ and the actual observed values of y. Allows us to estimate the line, even when x and y do not fall exactly on a line.

Understanding the regression line

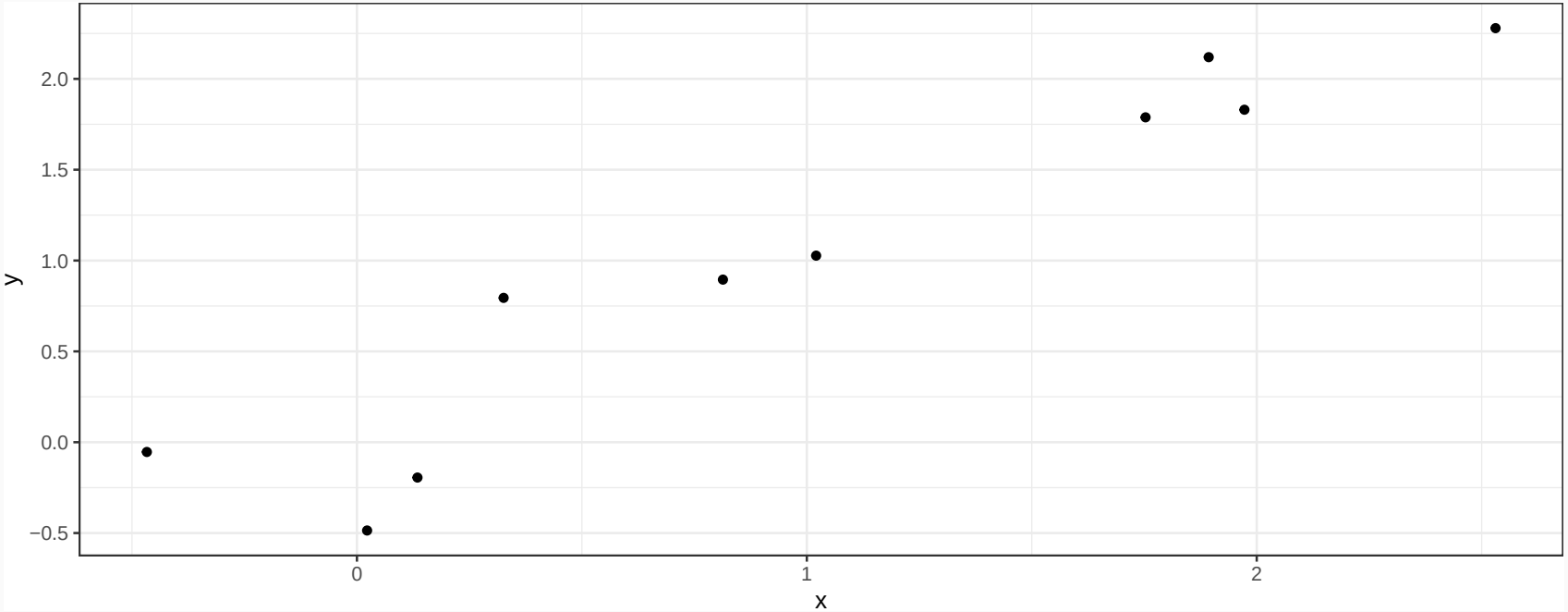
```
# A tibble: 10 x 2
         x       y
     <dbl>   <dbl>
 1  1.75    1.79
 2  0.0226 -0.486
 3  0.814   0.895
 4  2.53    2.28
 5 -0.467  -0.0537
 6  0.326   0.795
 7  0.135  -0.194
 8  1.97    1.83
 9  1.89    2.12
10  1.02    1.03
```

$\hat{\beta}_0 = 0.05$, $\hat{\beta}_1 = 0.95$

- Estimate $\hat{Y}$. Recall that $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$
- Estimate ε. Recall that $\varepsilon = Y - \hat{Y}$

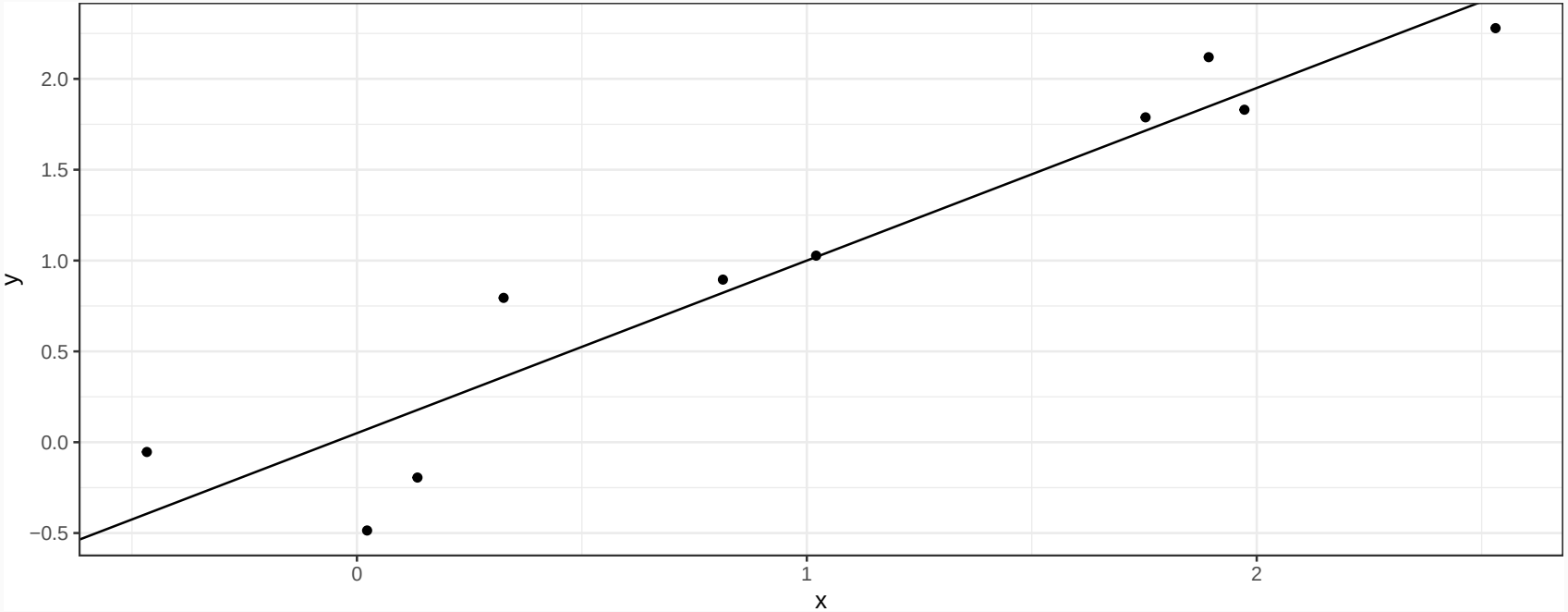
Understanding the regression line

$\hat{\beta}_0 = 0.05, \hat{\beta}_1 = 0.95$



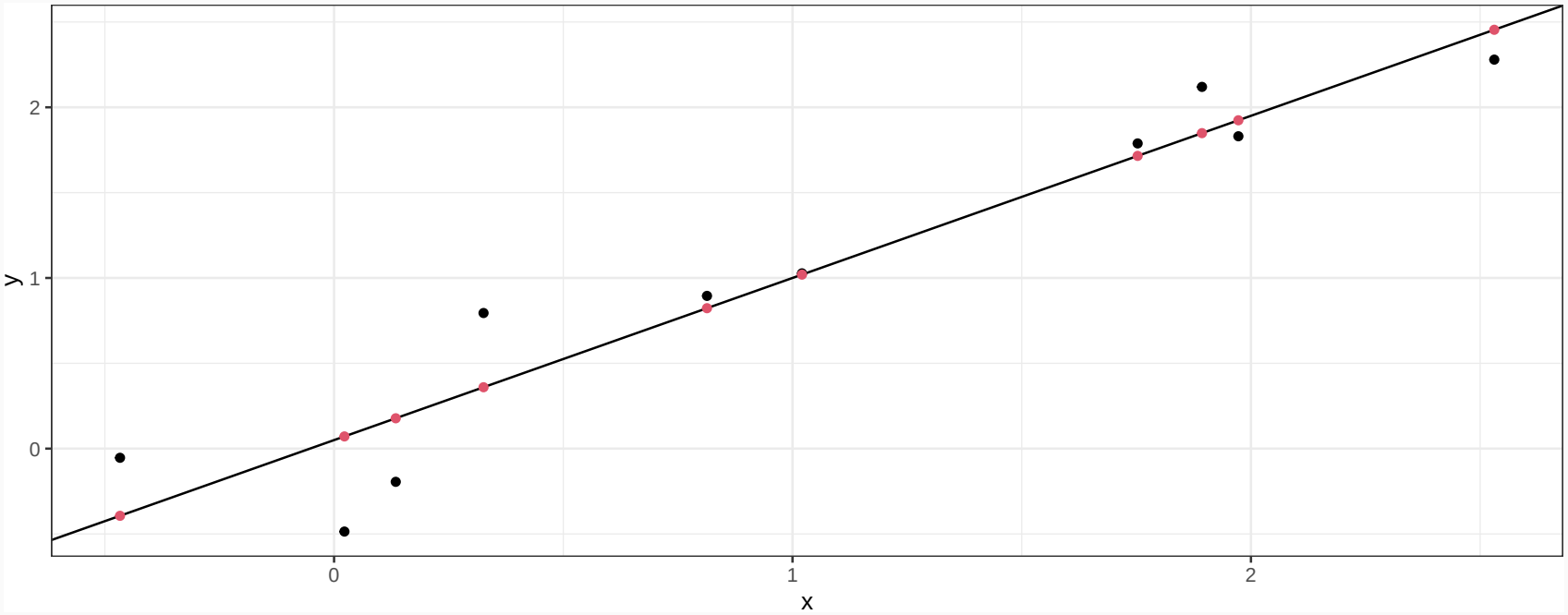
Understanding the regression line: adding the fit

$\hat{\beta}_0 = 0.05, \hat{\beta}_1 = 0.95$



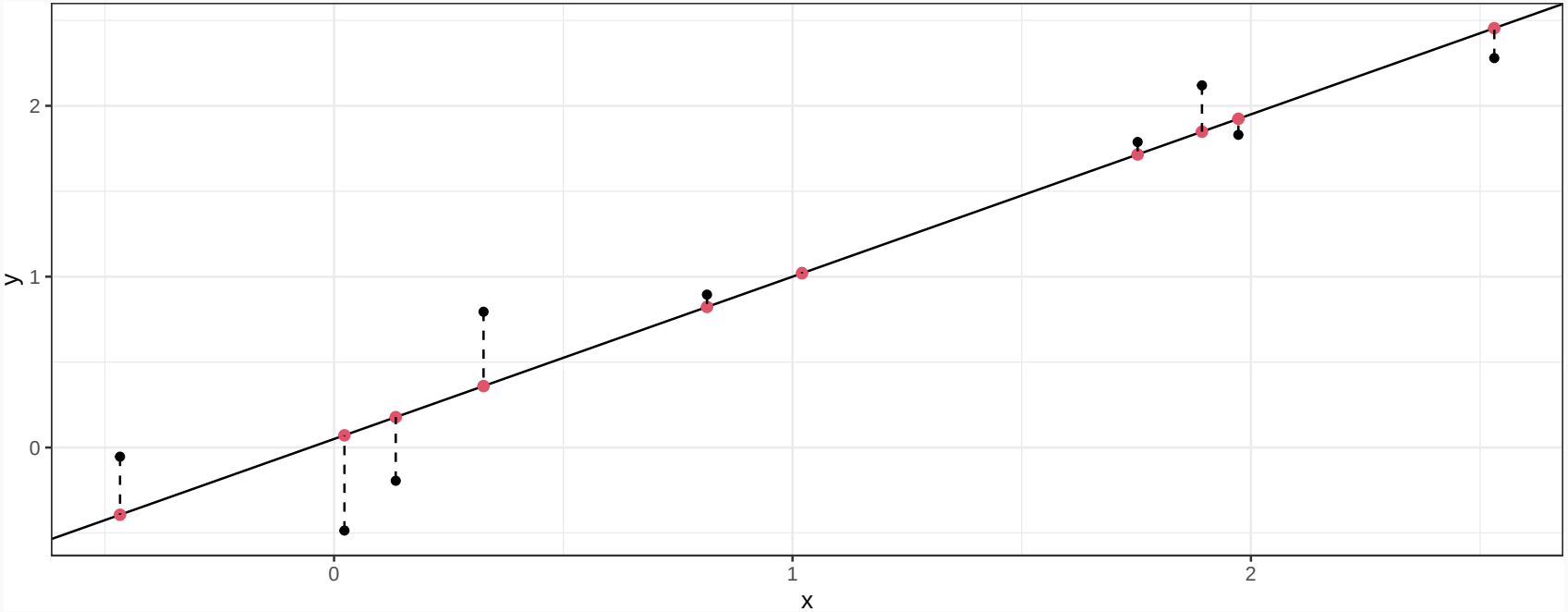
Understanding the regression line: adding $\hat{y}$

$\hat{\beta}_0 = 0.05, \hat{\beta}_1 = 0.95$



Understanding the regression line: adding ε

$\hat{\beta}_0 = 0.05, \hat{\beta}_1 = 0.95$



Assumptions of a linear regression model

For estimates of β to be unbiased and consistent, the following assumptions must be met:

1. The linear model approximates the data generating process

Assumptions of a linear regression model

For estimates of β to be unbiased and consistent, the following assumptions must be met:

1. The linear model approximates the data generating process
2. $E(\varepsilon) = 0$, errors uncorrelated with predictors

Assumptions of a linear regression model

For estimates of β to be unbiased and consistent, the following assumptions must be met:

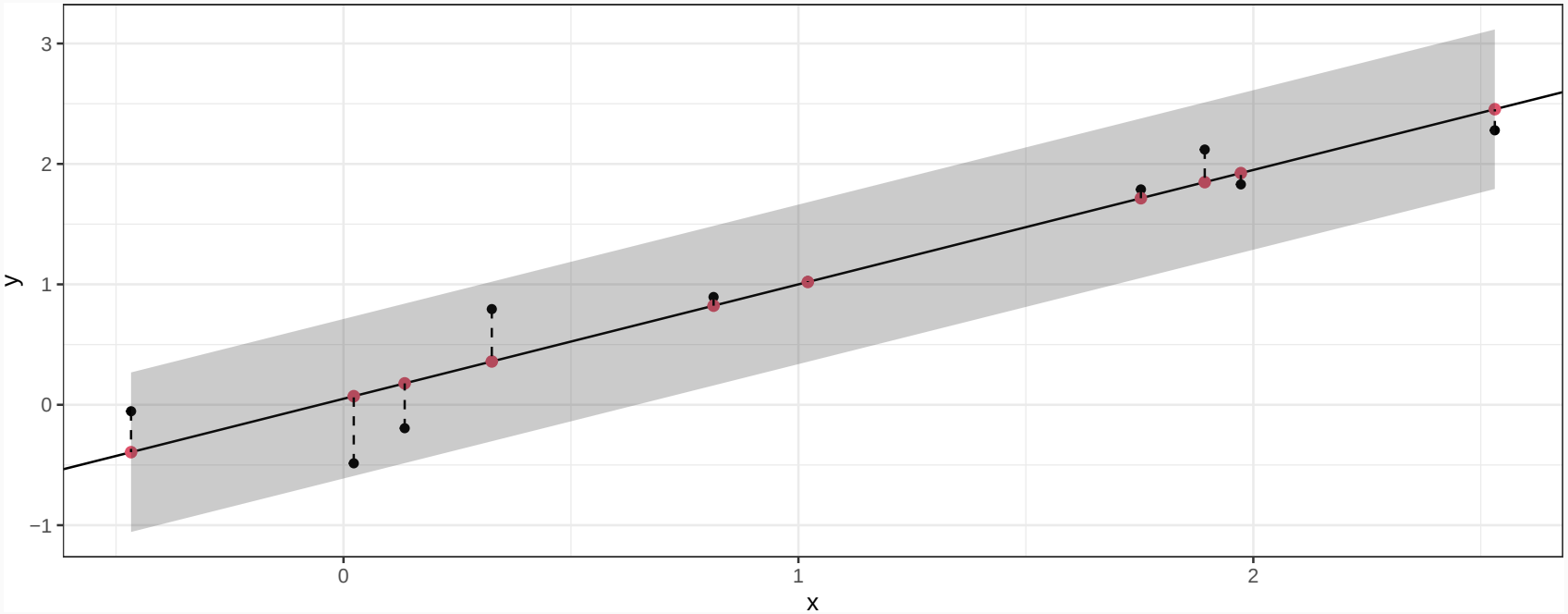
1. The linear model approximates the data generating process
2. $E(\varepsilon) = 0$, errors uncorrelated with predictors
3. Linear independence of predictors

Assumptions of a linear regression model

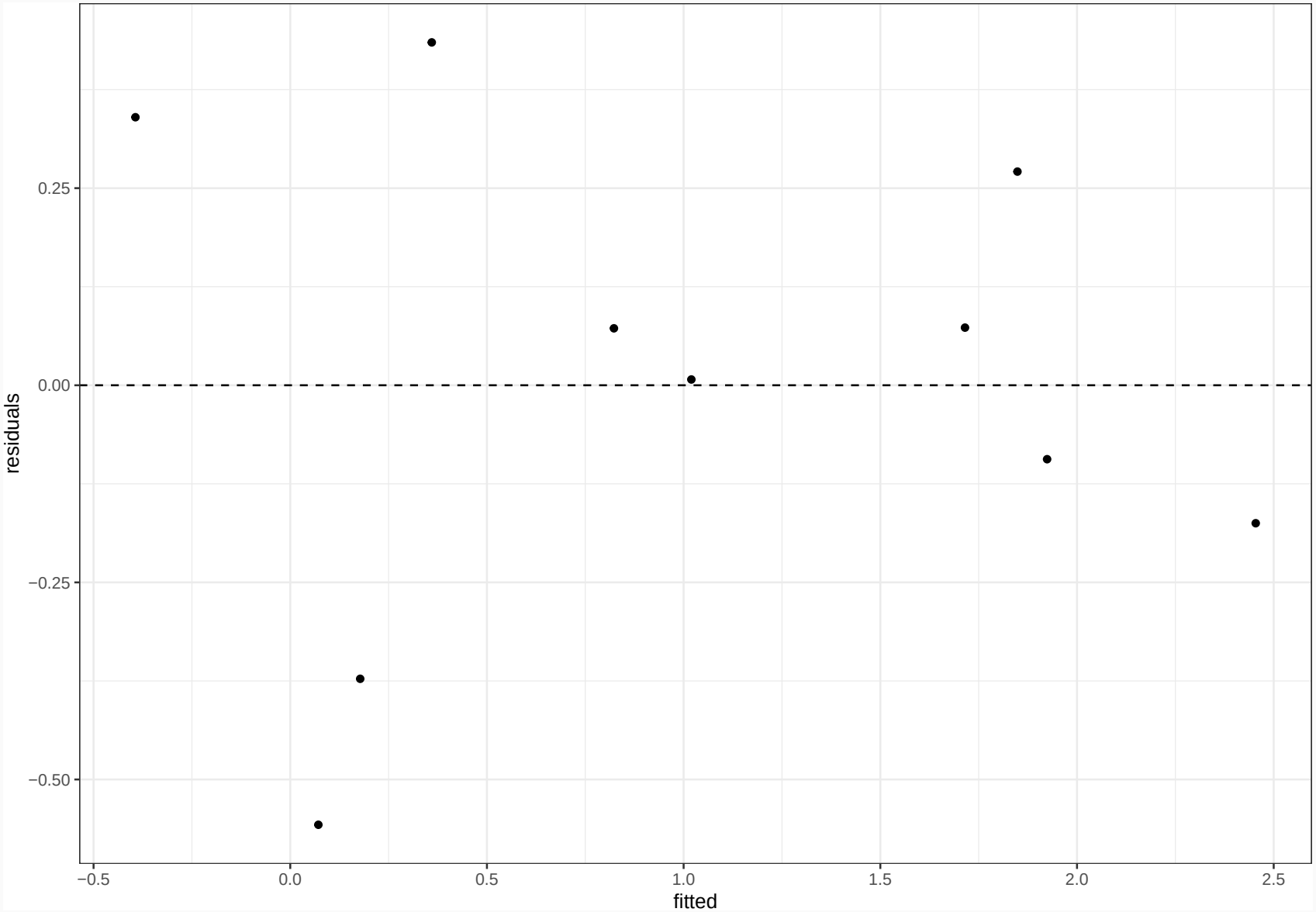
For estimates of β to be unbiased and consistent, the following assumptions must be met:

1. The linear model approximates the data generating process
2. $E(\varepsilon) = 0$, errors uncorrelated with predictors
3. Linear independence of predictors
4. Constant error variance (Homoskedasticity): $V(\varepsilon|X) = V(\varepsilon)$

What this means: 95% of observations should fall in this zone, regardless of location in x



One way to visualize: residuals vs fitted

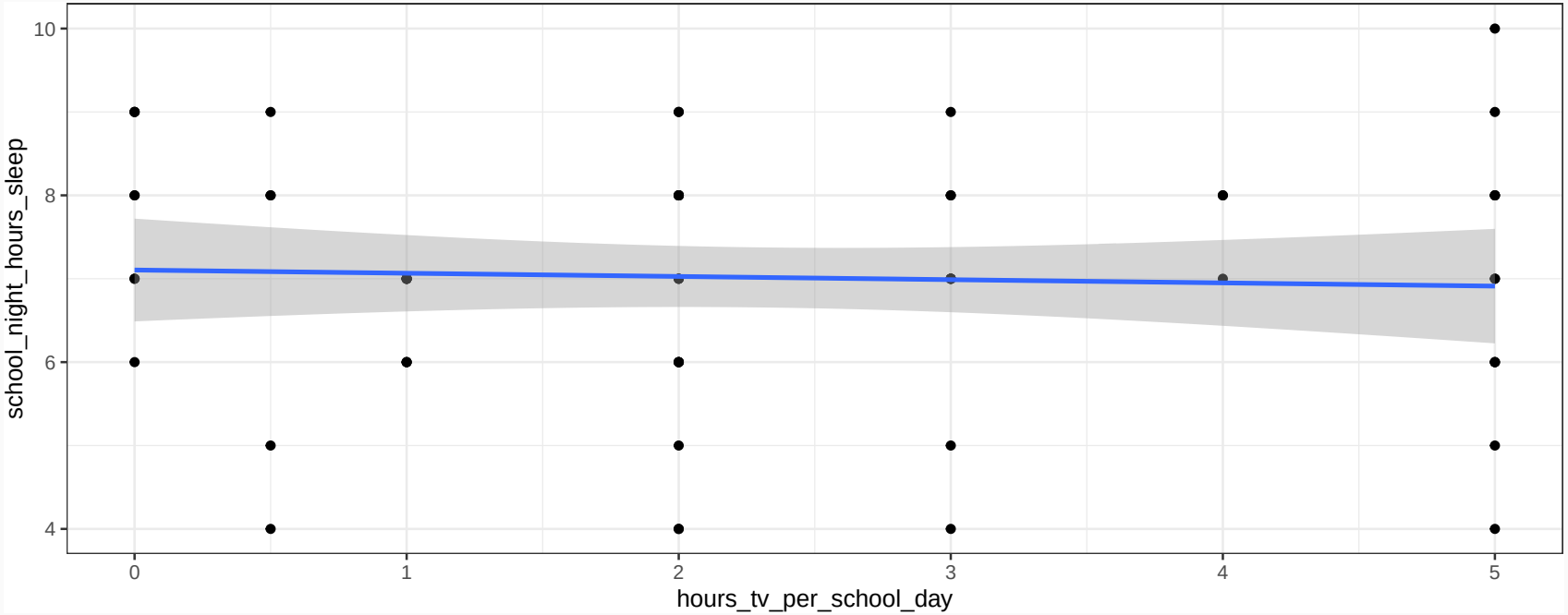


Let’s try this with real data

```
# A tibble: 64 x 13
     age gender grade hispanic race                    height weight helmet_12m
   <dbl> <chr>  <dbl> <chr>    <chr>                     <dbl>  <dbl> <chr>
 1    16 female    11 not      Black or African Americ~   1.5    52.6 never
 2    17 male      11 not      White                      1.78   74.8 rarely
 3    17 male      11 not      White                      1.75  107.  never
 4    18 male      12 not      Black or African Americ~   1.7    80.3 never
 5    16 male      10 not      White                      1.78   81.6 always
 6    16 male      10 not      Black or African Americ~   1.63   56.7 never
 7    14 male       9 hispanic White                      1.63   54.4 never
 8    17 male      11 not      Black or African Americ~   1.83   92.5 never
 9    15 male       9 not      Black or African Americ~   1.69   67.8 did not r~
10    17 male      10 not      White                      1.78   66.7 never
# i 54 more rows
# i 5 more variables: text_while_driving_30d <chr>, physically_active_7d <dbl>,
#   hours_tv_per_school_day <dbl>, strength_training_7d <dbl>,
#   school_night_hours_sleep <dbl>
```

Visualize our focal variables: Sleep and tv watching

```
ggplot(dat, aes(x = hours_tv_per_school_day, y = school_night_hours_sleep)) + geom_point() +
    geom_smooth(method = "lm")
```



Fit a model for sleep duration predicted by tv watching

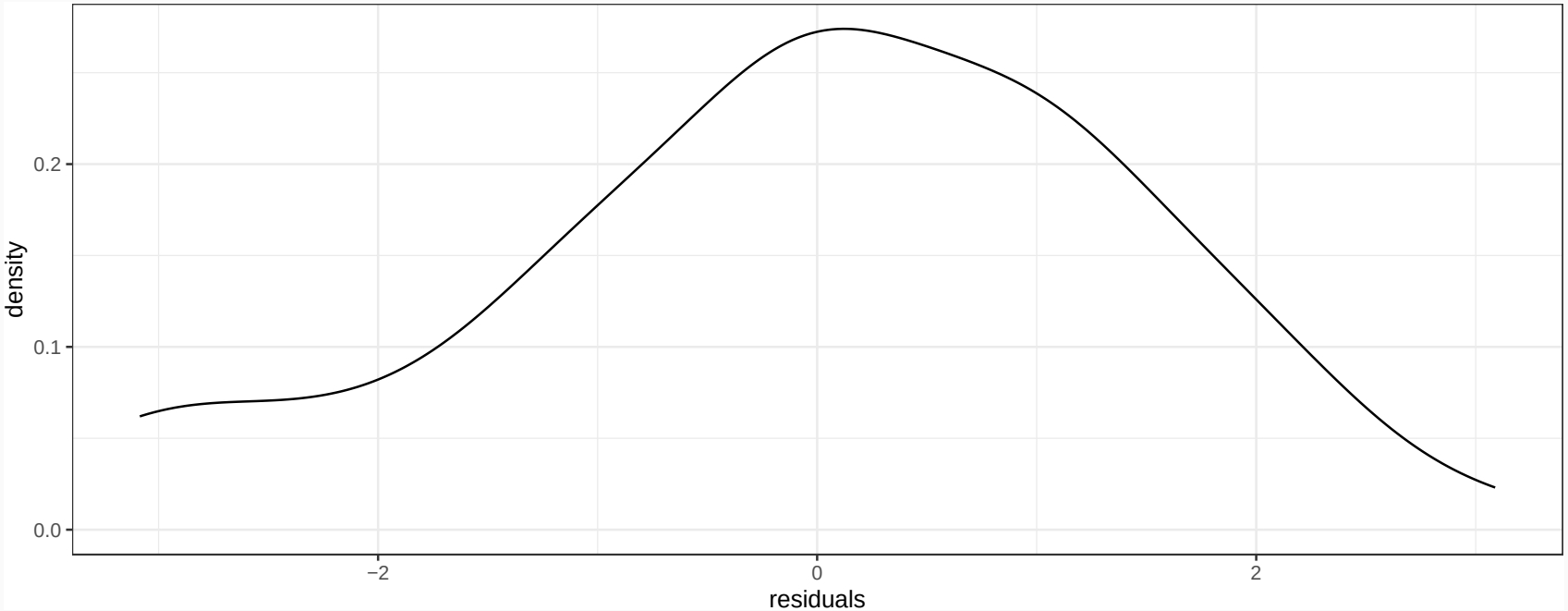
```
m1 <- lm(school_night_hours_sleep ~ hours_tv_per_school_day, data = dat)

tidy(m1)
```

```
# A tibble: 2 x 5
  term                    estimate std.error statistic  p.value
  <chr>                      <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)               7.10       0.308    23.1   4.15e-32
2 hours_tv_per_school_day  -0.0386     0.109    -0.356 7.23e- 1
```

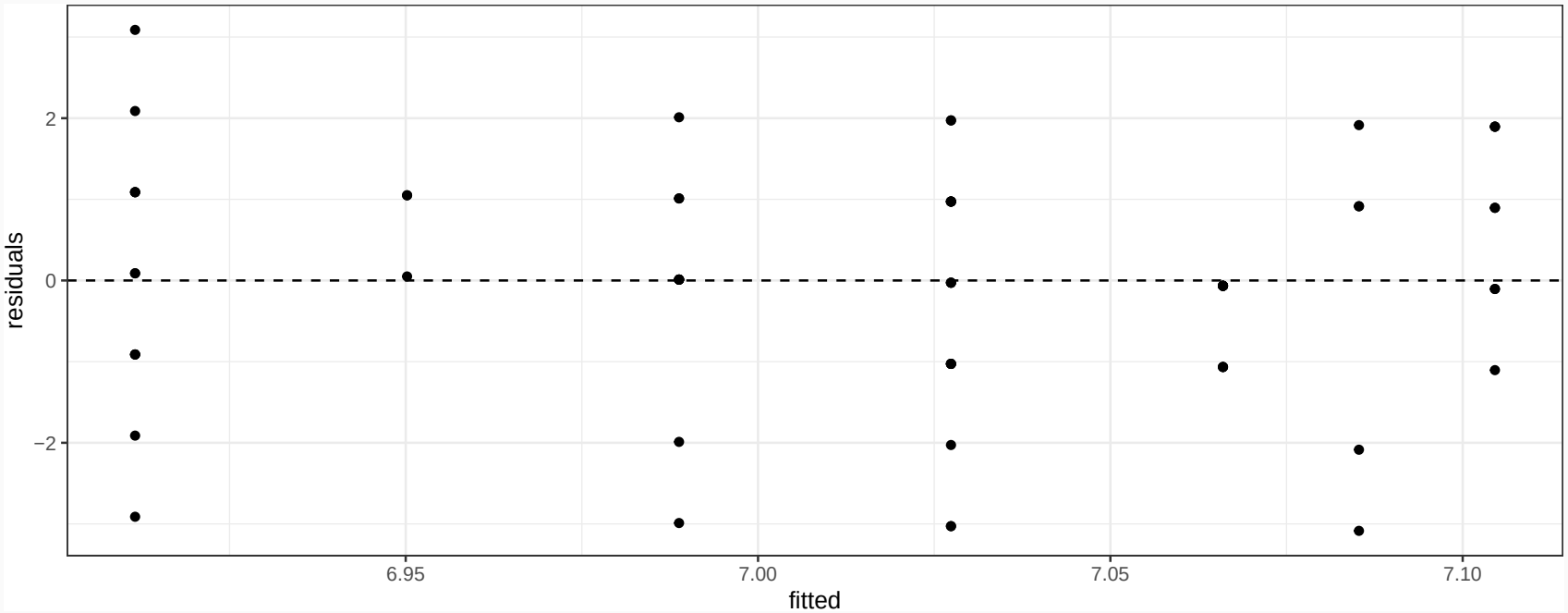
What is the distribution of the residuals? Are they Normal?

```
ggplot(data = data.frame(residuals = resid(m1)), aes(x = residuals)) + geom_density()
```



What about residuals vs fitted?

```
ggplot(data = data.frame(fitted = fitted(m1), residuals = resid(m1)), aes(x = fitted,
    y = residuals)) + geom_point() + geom_hline(yintercept = 0, lty = 2)
```



Sources of random variation in our model

$$\hat{\beta}_0 \sim N(\beta_0, s^2_{\beta_0})$$

$$\hat{\beta}_1 \sim N(\beta_1, s^2_{\beta_1})$$

$$y = \beta_1 + \beta_2 + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

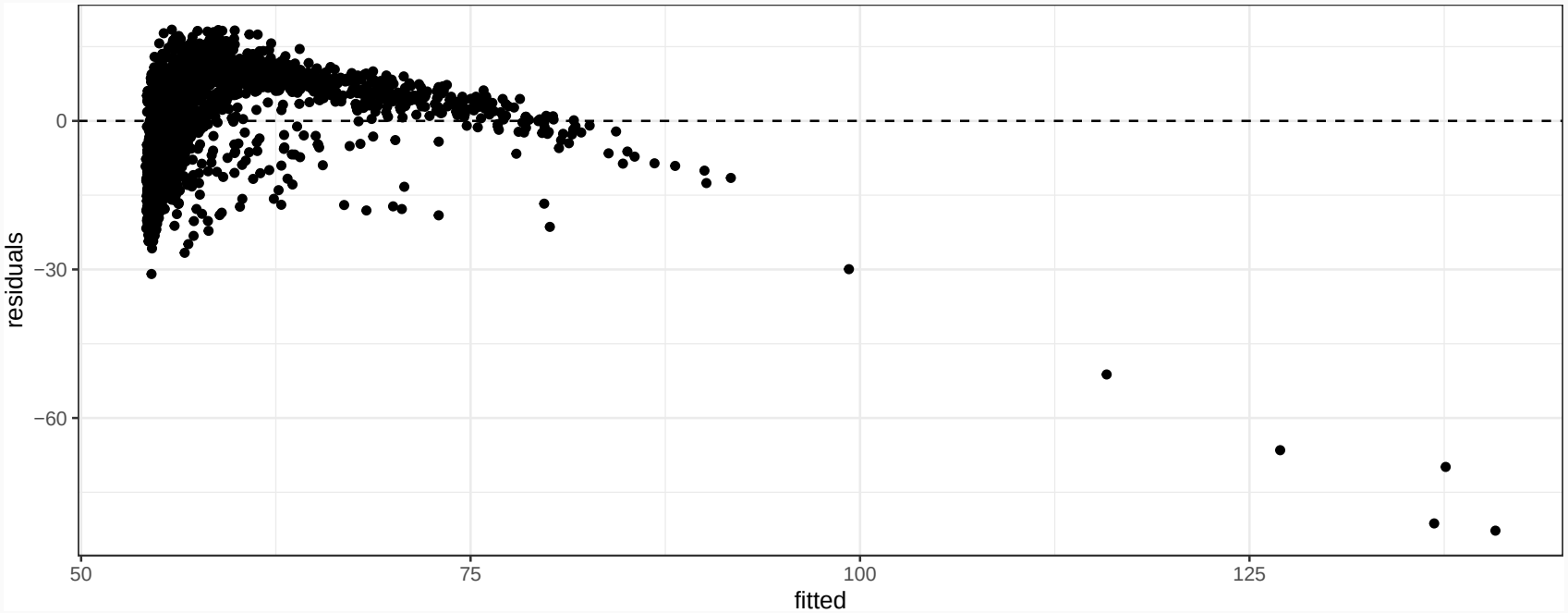
What does non-constant error variance look like?

Data with non-constant error variance will show distinctive patterns in their residuals

```
library(gapminder)
m2 <- lm(lifeExp ~ gdpPercap, data = gapminder)
```

Non-contant error variance

```
ggplot(data.frame(fitted = fitted(m2), residuals = resid(m2)), aes(x = fitted, y = residuals)) +
    geom_point() + geom_hline(yintercept = 0, lty = 2)
```



Problems

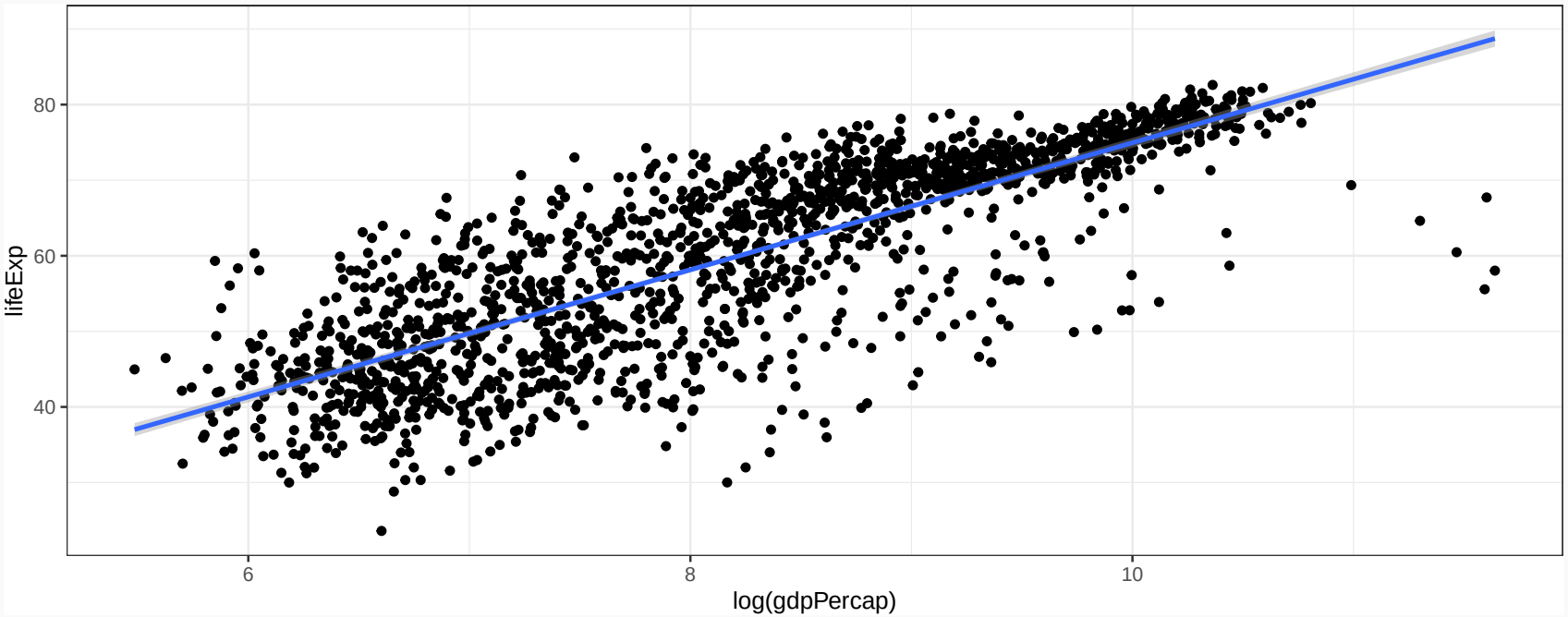
Here there are actually two problems:

1. The linear model does not reflect the data generating process (the relationship between x and y)
2. Error variance is not constant

OK, so log it

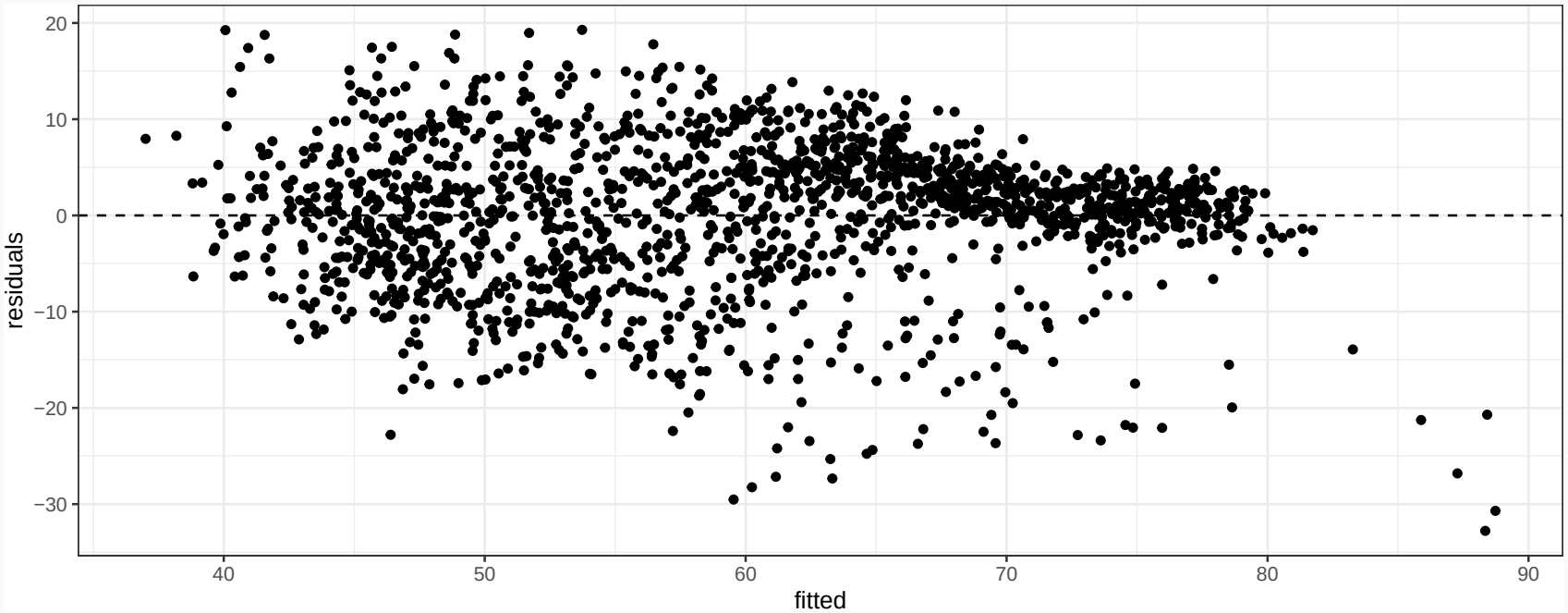
```
library(gapminder)
m3 <- lm(lifeExp ~ log(gdpPercap), data = gapminder)

ggplot(gapminder, aes(x = log(gdpPercap), y = lifeExp)) + geom_point() + geom_smooth(method = "lm")
```



Non-constant error variance

```
ggplot(data.frame(fitted = fitted(m3), residuals = resid(m3)), aes(x = fitted, y = residuals)) +
    geom_point() + geom_hline(yintercept = 0, lty = 2)
```



In summary

In linear regression, we assume that the residuals follow a normal distribution.

In summary

In linear regression, we assume that the residuals follow a normal distribution.

We can write this two ways

$$y = \beta_0 + \beta_1 X + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

We can write this two ways

$$y = \beta_0 + \beta_1 X + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

Or

$$y \sim N(\mu, \sigma^2)$$

$$\mu = \beta_0 + \beta_1$$

Two kinds of uncertainty: y and mu

Let’s return to the gapminder model

```
m3 <- lm(lifeExp ~ log(gdpPercap), data = gapminder)
summary(m3)
```

```
Call:
lm(formula = lifeExp ~ log(gdpPercap), data = gapminder)

Residuals:
    Min      1Q  Median      3Q     Max
-32.778  -4.204   1.212   4.658  19.285

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     -9.1009     1.2277  -7.413 1.93e-13 ***
log(gdpPercap)   8.4051     0.1488  56.500  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.62 on 1702 degrees of freedom
Multiple R-squared:  0.6522,    Adjusted R-squared:  0.652
F-statistic:  3192 on 1 and 1702 DF,  p-value: < 2.2e-16
```

Uncertainty in mu

We have *uncertainty* in both the location of the intercept and slope

$$\hat{\beta}_0 \sim N(\beta_0, s_{\beta_0})$$

$$\hat{\beta}_1 \sim N(\beta_1, s_{\beta_1})$$

Uncertainty in mu

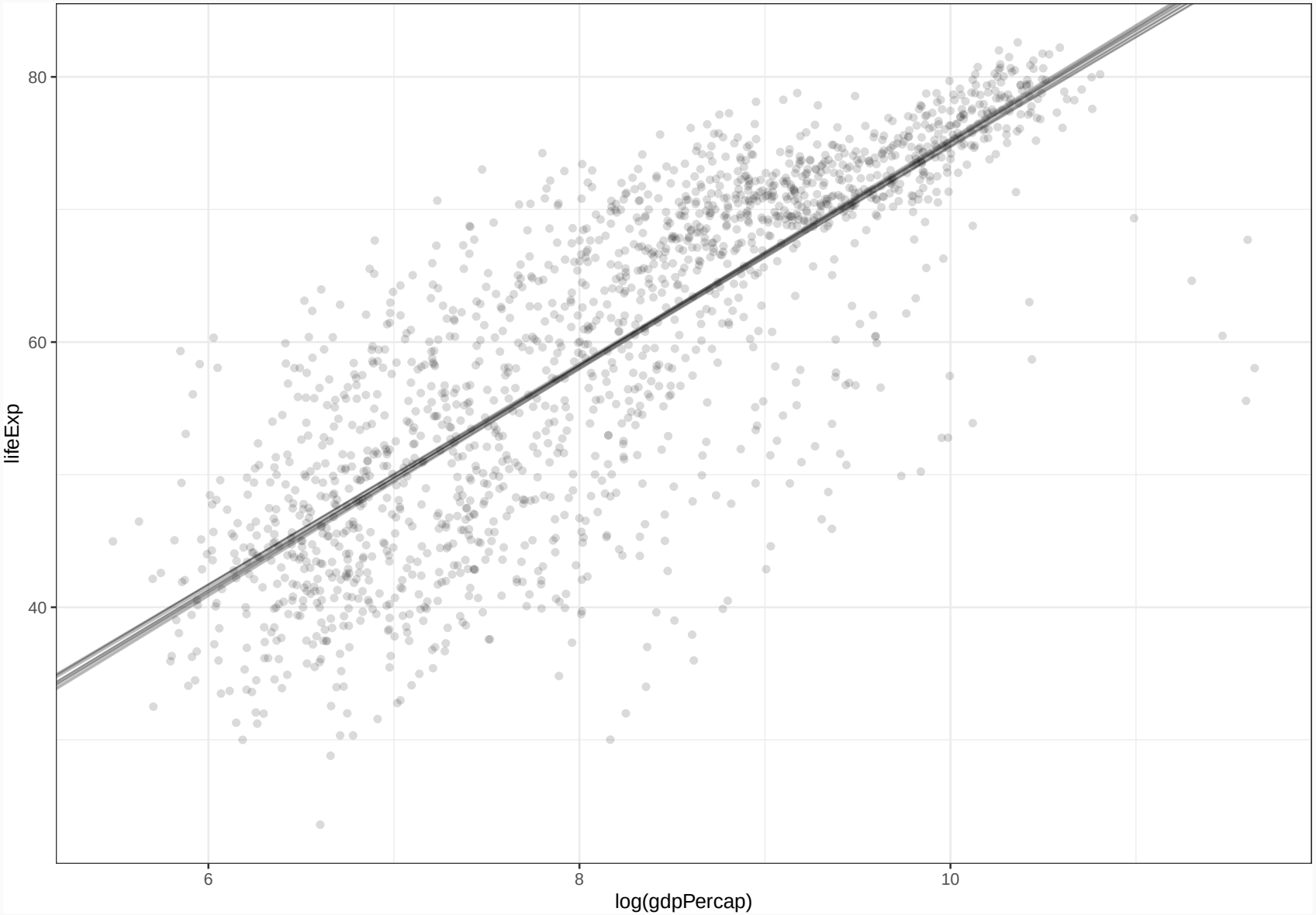
We have *uncertainty* in both the location of the intercept and slope

$$\hat{\beta}_0 \sim N(\beta_0, s_{\beta_0})$$

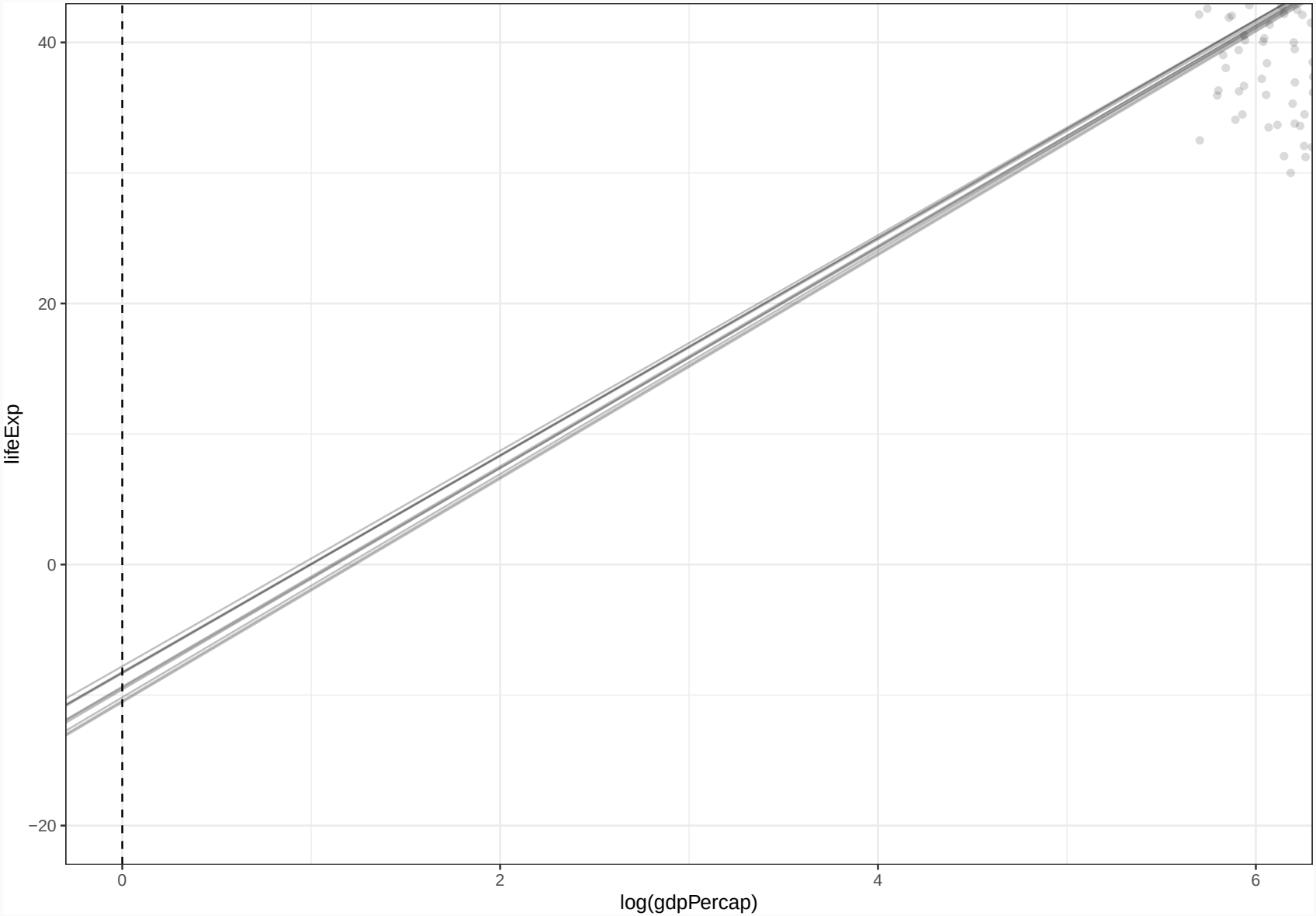
$$\hat{\beta}_1 \sim N(\beta_1, s_{\beta_1})$$

What does this uncertainty mean?

Many possible regression lines

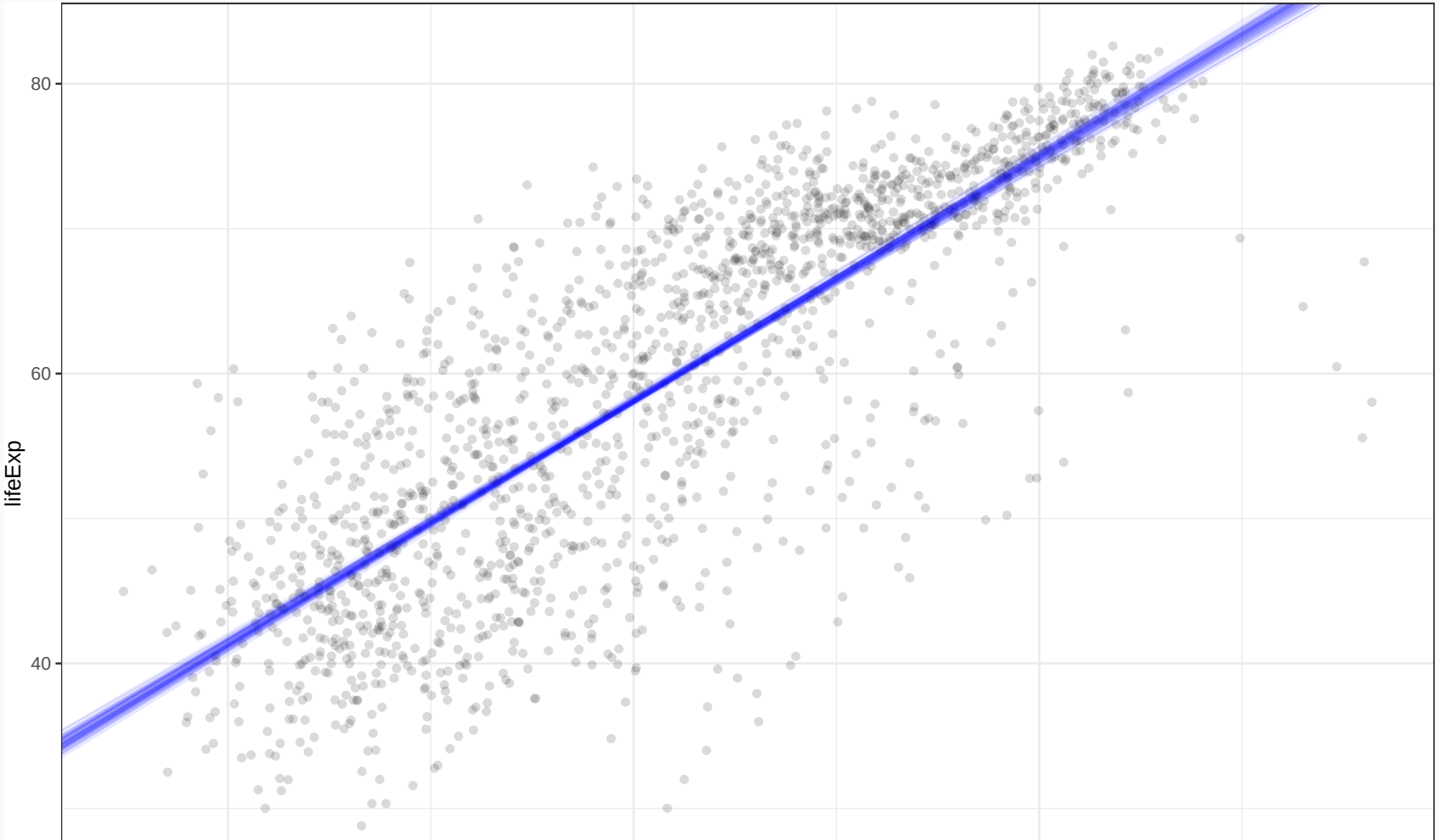


Zooming in on intercept and slope uncertainty



Visualizing uncertainty in mu

```
post <- as_draws_df(m3_b) |>
    rename(a = "(Intercept)", b = "log(gdpPercap)") |>
    slice_sample(n = 100)
ggplot(gapminder, aes(x = log(gdpPercap), y = lifeExp)) + geom_point(alpha = 0.15) +
    geom_abline(data = post, color = "blue", aes(intercept = a, slope = b, group = .draw),
        alpha = 0.05)
```



Adding uncertainty in y

We get uncertainty in mu by evaluating how the sampling distributions of β work. On top of this uncertainty, we have uncertainty in y, driven by $sigma$

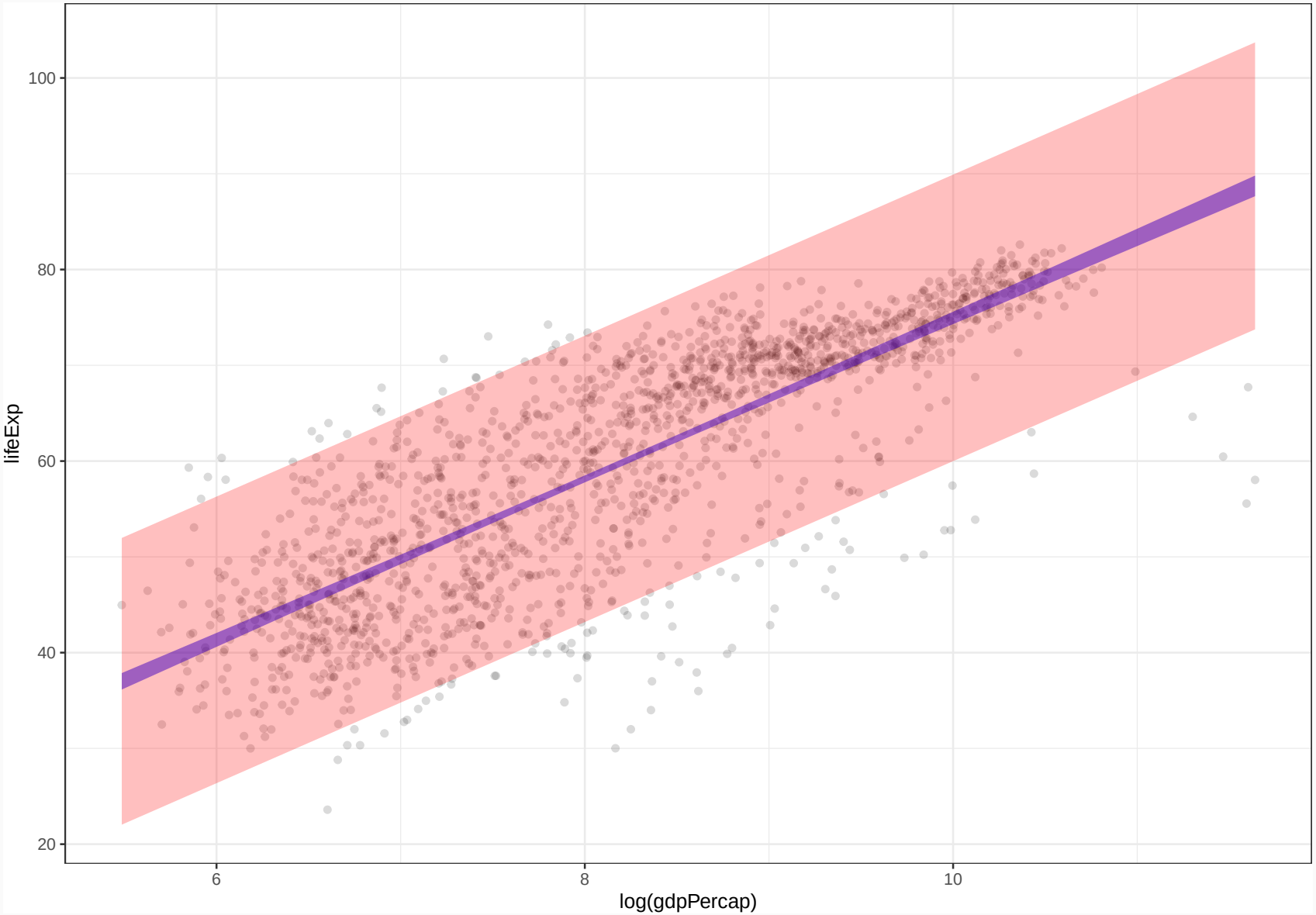
```
mu_uncertainty <- predict(m3, interval = "confidence")
sampling_uncertainty <- predict(m3, interval = "prediction")
head(mu_uncertainty)
```

```
       fit      lwr      upr
1 46.86506 46.29702 47.43310
2 47.30012 46.74364 47.85661
3 47.62400 47.07600 48.17200
4 47.45579 46.90339 48.00818
5 46.42835 45.84855 47.00816
6 46.93666 46.37053 47.50279
```

```
head(sampling_uncertainty)
```

```
       fit      lwr      upr
1 46.86506 31.90963 61.82050
2 47.30012 32.34512 62.25512
3 47.62400 32.66931 62.57869
4 47.45579 32.50094 62.41064
5 46.42835 31.47247 61.38424
6 46.93666 31.98130 61.89202
```

Visualizing



In summary

We have multiple sources of random error in our model.

1. We have uncertainty in our estimates of β, that we approximate using the Central Limit Theorem. This is the uncertainty we have about the location of $E(y)$ or μ
2. We have uncertainty in y, which is ordinary sampling error. For linear regression, we are going to assume that conditional on X, this error is Normally distributed